

CHAPTER 2

LITERATURE REVIEW

In this chapter, the literature review will help in gathering the information and find the suitable solution for the problem that is faced in this project. The analysis to be conducted will provide a lot of information about the domain involved. This information is very helpful to understand the domain being studied in more depth. In addition, appropriate techniques for solving the problems of this project can also be identified from the comparisons that will be made based on several criteria.

2.1 Internship Management and E-Logbook Systems

2.1.1 Internship

Internships are learning programs designed in such a way that learners get a chance to gain practical experience in an office or professional environment where they put into practice what they have learned theoretically in school. The internship program is very vital in connecting the institution with the industry because it helps students to understand the real working environment. According to Wan Abdul Rahman et al. (2024), internship programmes enable students to apply classroom knowledge in real working environments, thereby preparing them for future careers and improving their employment readiness. In addition, Bawica (2021) confirms that internship programmes significantly enhance students' employability readiness by exposing them to real workplace environments and improving their transition from academic to professional settings.

The role of internships is very important in higher education since they offer chances for acquiring competencies that cannot be acquired in the classroom environment alone. Internships allow learners to gain skills, improve their communication skills, and develop problem-solving skills relevant to their fields of study. In addition, internship programs allow universities to evaluate the success of their curriculum by assessing the performance of students in the working environment. Employers also benefit from internships, as such programmes provide companies with competent job applicants and contribute to the development of skilled graduates (Wan Abdul Rahman et al., 2024). Furthermore, Okoye and Edokpolor (2021)

emphasise that industrial work experience plays a significant role in developing employability skills such as teamwork, adaptability, and job-related competencies.

In addition to improving employability, internships encourage students to develop professional attitudes, workplace ethics, and teamwork skills (Wan Abdul Rahman et al., 2024). Students are required to complete assigned tasks, communicate with colleagues, and adapt to organisational cultures. According to Bawica (2021), structured internship experiences improve students' confidence and career readiness, while Okoye and Edokpolor (2021) highlight that experiential learning strengthens the transition from academic environments to the workforce. Therefore, internship programmes are considered an essential component of higher education that supports both academic and professional development.

2.1.2 Internship Management

Internship management refers to the process of planning, organising, monitoring, and evaluating internship activities involving students, academic supervisors, and industry supervisors. Management of internships helps ensure that students are able to achieve the desired learning outcomes, while at the same time allowing supervisors to track their progress. Internship management involves various stages such as internship placement, internship activities management, performance assessment, communication, and document management (Wan Abdul Rahman et al., 2024).

The increasing number of students participating in internship programmes has highlighted the need for efficient internship management practices. There is a need for academic institutions to coordinate many activities such as internship enrolment, supervision allocation, monitoring of interns' performance, report submission, and evaluation. Without proper management processes, information regarding internship can be dispersed, causing difficulties in performance monitoring and stakeholder coordination (Ahmed et al., 2026). Recent studies also emphasise the importance of digital transformation in internship administration, particularly through integrated cloud-based systems that improve coordination and data accessibility (Wan Abdul Rahman et al., 2024).

According to Wan Abdul Rahman et al. (2024), centralised, cloud-based internship management systems can improve the efficiency of internship administration by consolidating communication, attendance tracking, and logbook documentation into a single platform. Such systems enable students to access internship-related information more easily while allowing supervisors to monitor student progress and provide timely feedback. In addition, Setiaji et al. (2025) highlight that cloud-native internship management platforms further enhance system efficiency by improving integration, scalability, and overall workflow automation through modern system architecture. Therefore, effective internship management plays a crucial role in ensuring a successful internship experience for students and educational institutions.

2.1.3 Current Issues in Internship Management

Management of internships is a very important aspect of tertiary education, although there are many problems which continue to exist, thus making this process ineffective. One of them is the continuous reliance on the old system of logbooks where information about students' everyday activities during their internships is recorded. Despite the simplicity of this method, these records are easily lost and damaged. According to Ahmed et al. (2026), this method also limits accessibility, making it difficult for supervisors to review students' progress promptly, especially when internships take place far from the institution. Similarly, Seman et al. (2022) highlight that traditional logbook systems often lead to inefficiencies in monitoring student progress and create delays in supervisor evaluation due to lack of real-time access.

In addition to documentation issues, communication gaps between students, academic supervisors, and industry supervisors present another significant challenge. The absence of a centralised platform often results in delayed feedback, lack of coordination, and unclear expectations, particularly when companies do not provide regular updates to the institution (Wan Abdul Rahman et al., 2024). Furthermore, manual record-keeping can lead to incomplete entries, illegible handwriting, and inconsistent reporting, particularly when students delay updating their logbooks. These factors reduce the accuracy and reliability of internship records (Ahmed et al., 2026). Petcharat and Sutthirak (2024) further support that inefficient communication and fragmented systems can significantly affect internship management effectiveness, especially in coordinating stakeholders across different organisations.

In addition, the supervision process itself is fraught with difficulties for administrators who have to assess interns' performance in writing (Ahmed et al., 2026). Managing paper-based records also reduces the efficiency of data storage and retrieval. Consequently, many educational institutions have adopted digital internship management systems to support real-time monitoring, improve communication, and simplify administrative processes, thereby enhancing the overall effectiveness of internship programmes. Petcharat and Sutthirak (2024) further emphasise that web-based and mobile-enabled internship systems improve supervision efficiency, reduce manual workload, and support better tracking of student performance throughout the internship period.

2.2 Information Systems for Internship Management

2.2.1 Internship Management Systems

Internship Management System (IMS) is a type of information system that is developed with the purpose of supporting the management, monitoring, and evaluation of internship programs. It serves as a central portal where internship related information is accessible by both students and academic as well as industrial supervisors, who can also exchange communication through it. The ultimate goal of developing an internship management system is to make the process of internship administration more efficient.

The need for internship management systems has become more crucial as internships become increasingly complex within institutions of higher learning. The conventional methods of administrating these programs involve numerous modes of communications and documentation. As noted by Thammastitkul (2023), the use of a cloud-based internship management system makes the process more effective through the integration of communication, documentation, and monitoring into one system. This leads to the minimization of confusion among students as well as availability of internship related information and resources. Moreover, Ambara et al. (2024) highlight that inefficient documentation and supervision in conventional internship management make the entire process less efficient.

In addition, management systems for internships facilitate better observation of the activities undertaken by students and help give feedback in a timely manner if needed.

Centralized internship information makes it easier to make decisions and adds to the process of better organization of internship management. Sidik et al. (2024) point out the importance of real-time monitoring systems that increase the efficiency of supervision and communication between students and their supervisors and decrease delays in performance evaluation and reporting. Thus, the use of IMS is very important for improving internship management.

2.2.2 Mobile-Based Internship Systems

Mobile-based internship systems are applications designed to allow users to manage internship activities through smartphones and tablets. The widespread adoption of mobile devices has encouraged educational institutions to utilize mobile applications as a means of improving accessibility and user convenience. Mobile systems allow students to access internship information, submit reports, record activities, and communicate with supervisors regardless of location.

According to Castro (2024), mobile-based internship systems are accessible anytime and anywhere on Android devices, allowing users to review their progress and related information without dependence on desktop computers or physical paperwork. This flexibility is particularly beneficial for internship students who frequently operate outside university environments, allowing them to document activities immediately after completing assigned tasks. Castro (2024) further developed a mobile-based internship monitoring system that incorporates a GPS locator to verify trainees' presence at their assigned work locations, enabling students to record daily activities and track internship progress through a centralized platform. The study demonstrated that mobile technologies improve monitoring efficiency and simplify internship documentation processes. Therefore, mobile-based internship systems provide significant advantages in terms of accessibility, convenience, and real-time information management.

In addition, Wibowo et al. (2024) highlight that Android-based internship management systems enhance real-time monitoring and supervision by enabling continuous tracking of student activities through mobile platforms. Such systems improve communication between students and supervisors while ensuring that internship documentation is updated efficiently and accurately. Similarly, Kapoh et al. (2024) emphasise that digital internship management

systems support real-time monitoring and centralized data management, which significantly improves administrative efficiency and reduces delays in supervision and reporting processes.

2.2.3 Cloud-Based Internship Systems

Cloud computing is a vital part of today's information systems owing to its capability to provide centralized storage of information as well as remote access to that information. Cloud internship management system utilizes cloud computing technologies for storing and managing information related to internships via servers connected through the Internet.

As Setiaji et al. (2025) indicate, an intern management platform based on cloud technology enables real-time monitoring, automatic registration of attendance, centralized evaluation of performance, and scalability, role-based access control, and security through data encryption. When applied in educational institutions, cloud technologies facilitate the exchange of information between students, lecturers, and administrators and also reduce the need for infrastructure. Furthermore, as indicated by Li et al. (2024), cloud computing facilitates real-time monitoring and centralized processing of data.

Similarly, Thammasitkul (2023) pointed out that technologies of cloud computing enhance the management of internships owing to their capability to provide centralised data storage as well as real-time access to the internships' data. Additionally, cloud computing systems offer automatic backup and synchronisation capabilities that help avoid data loss. Furthermore, Li et al. (2024) state that cloud computing architecture allows for scalable data management as well as multi-device data accessibility, thus making it possible to access and update data on mobile and web platforms.

2.2.4 Role of Information Systems in Internship Management

Information systems have an important part to play in improving the effectiveness of the management of internships. They help with communications, monitoring, documenting, and administration of the process of internship. In a central digital environment, it becomes easy for both students and supervisors to access information related to internship, submit reports, monitor their progress, and communicate with each other effectively. The first of the

main contributions of information systems is the improvement in communication between the parties involved. Proper communication will allow students to get continuous guidance during the time of internship.

Moreover, information systems help in internship supervision and evaluation. The supervisors will be able to check the work that students submit, and keep track of the internship progress and possible problems that can arise with greater ease than the manual system. The digital record keeping system makes it easier to manage internships through structured storage of the information. As mentioned by Thammastitkul (2023), central internship management systems provide better coordination and efficiency. Thus, information systems are important tools in modern internship management.

The significant advancement in the field of information systems is the creation of Digital E-Logbook systems that are aimed at replacing the old logbooks made on papers. This application permits the students to record their internships activities electronically using either mobile or internet-based software, thereby ensuring centralized data storage for all internship logs. The usage of digital logs has greatly increased owing to the advancement in mobile technology and cloud computing. In one example, Castro (2024) created a mobile-based internship log that enables students to record their daily activities and monitor their performance easily. Unlike the old manual way, using digital E-Logbook, students can edit their logs anywhere and at any time, whereas the supervisor does not have to collect physical logs.

However, despite all these benefits, there are internship management systems which still use manual or partially digital means, according to earlier research findings (Ahmed et al., 2026; Dewi & Hanifah, 2024). The majority of current systems concentrate mainly on the aspects of activity logging and monitoring. They lack features for improving students' documentation work quality. These features can include AI writing help, automated reminder notifications, image documentation, intelligent feedback, among others. Moreover, these features are created separately rather than being integrated in a single system. It demonstrates a research gap in the development of the internship management system with digital logbook features along with other useful capabilities. That is why the E-Logbook Mobile Application will be developed to fill the research gap.

2.3 Artificial Intelligence and Smart Technologies

2.3.1 Artificial Intelligence in Education

Artificial Intelligence is defined as computer systems that are able to perform activities that need human intelligence in order for them to be performed. Such activities include learning, reasoning, problem solving, decision making, and language understanding. AI technologies have received much attention in many areas including the education sector.

In educational settings, AI systems are employed to enhance learning processes, tailor learning content, and help with academic endeavors. AI systems can help students by giving suggestions, creating learning materials, and providing feedback in response to user input. According to Lang et al. (2024), due to the development of large language models, modern AI systems have been able to provide much more sophisticated features including content creation, writing assistance, and communication in natural language. Likewise, Mah and Groß (2024) stated that the use of AI in higher education has made teaching and learning more efficient through educational productivity enhancement, faculty support, and intelligent technologies integration at universities. Moreover, Peláez-Sánchez et al. (2024) stated that large language models had become an integral part of Education 4.0 because of the ability of AI to help with learning, creation of academic content, and intelligent communication.

The adoption of AI technology in educational systems presents possibilities of making education more productive and less of a burden administratively. With the continued use of technological innovations in education, AI has become one of the major elements in the construction of intelligent information systems

2.3.2 AI-Assisted Writing

AI-assisted writing involves the application of artificial intelligence technologies to enable the users to produce, enhance, and refine their texts. These tools make use of natural language processing and machine learning methods in analyzing the text and making suggestions on issues such as grammar, spellings, sentence construction, readability, and structuring of text.

Good writing ability is one of the factors that are critical for documenting internships as interns are expected to write about their activities and experiences. Not all the students have good writing abilities, and that could affect how they document their internships. One of the benefits of using AI writing assistant tools is that the students get assistance and automated suggestions while they are doing their writing. Mah and Groß (2024) indicated that AI-based technologies are becoming popular in academic work in institutions of higher education, including writing, feedback and content creation.

According to Lang et al. (2024), large language models like Gemini and GPT-4 have the potential of supporting users in producing better-written materials. They can facilitate students in crafting better logbooks with improved professionalism and structure while minimizing the mistakes made during writing. Moreover, Peláez-Sánchez et al. (2024) mentioned that large language models have greatly supported academic writing through helping users in creating content, generating ideas, polishing languages, and encouraging interactive learning. AI writing assistance tools can be productive by saving time spent on documentation.

The integration of AI-assisted writing into internship management systems can therefore enhance the quality of student reports while supporting learning and professional development throughout the internship period.

2.3.3 Reminder and Notification Systems

Systems of reminders and notifications have been incorporated into various applications for their effective usage in improving user engagement and accomplishing tasks and providing information. These systems offer reminders and notifications related to important events, due dates, updates, and incomplete tasks. The use of timely reminders encourages users to accomplish their duties.

Educational settings benefit from reminders since they facilitate learners to control their academic undertakings like submitting assignments, monitoring attendance, and writing reports. Reminders may be provided via mobile apps, emails, or system notifications to make sure that the users are aware of any action that needs to be taken by them. Mustapa and Mustapa

(2023) noted that mobile academic reminder applications are beneficial to the users in managing their important deadlines and academic obligations.

In the case of internship management, there is an opportunity to use reminders for students to have consistent entries in their logbooks during the whole internship period. Instead of solely relying on their self-motivation, students will be receiving automated reminders which will prompt them to enter their activities and provide necessary documentation. As a result, there is less probability of incomplete entries and better accuracy of internship documentation. In addition, according to Osowski et al. (2023), personal reminders have been shown to positively influence the user's timeliness of task completion.

Effective reminder systems contribute to better time management, increased user participation, and improved consistency in task completion. Therefore, notification technologies are valuable components in modern internship management platforms.

2.3.4 Image Documentation Support

Image Documentation is defined as the use of images or visuals to back up information recorded and activities carried out. There is an increasing trend of using digital systems to facilitate the uploading of images for the sake of improving record keeping and providing supporting proof. Images could capture actions taken in the workplace, achievements accomplished, projects undertaken, and other internship experiences.

In internship management systems, image documentation acts as a proof of the activities done by students. The supervisors may be able to view the images and gain insight into the activities carried out by the students. This ensures that the documentation is more reliable.

In addition, image-based documentation helps in fostering reflective learning because it allows the learners to document their experiences and accomplishments during the internship period using visuals. By having written information combined with images, the internship documentation becomes complete and effective. According to Weber et al. (2023), ePortfolio-based internship system helps in documenting the learners' experience by allowing them to

submit multimedia documents, including images for reflection and evidence of the learning experiences in order to enhance the assessment process. Furthermore, according to Hartono (2024), web-based internship system improves the quality of documentation by recording daily activity and attaching evidence such as images.

2.4 Development Tools and Technologies

2.4.1 Flutter Framework

Flutter is a multiplatform mobile application development framework developed by Google, which allows developers to develop Android and iOS apps using the same code base. This framework contains a wide range of widgets, the concept of declarative user interface (UI), and the hot reload functionality that helps in developing mobile applications quickly and easily. Flutter has gained widespread acceptance in mobile application development because of its versatility, efficiency, and vibrant open-source community, and it allows development of native mobile, web, and desktop apps from a single code base (Kishore et al., 2022).

In Sharma et al. (2022), the authors explore hybrid development strategies in Flutter, paying attention to the widget system provided by Flutter and supporting the mixing of platform-native code with UI built with Dart. An analysis of Flutter's widget system, which is the basis for the declarative UI approach, helps understand why it is possible to support hot reloading and fast iterations without excluding the possibility of implementing platform-specific features using native modules. It can be especially valuable when speaking about the relevance of using Flutter in cases where developers will have to iterate over the UI more than implement the native part.

While Flutter provides built-in native integration for both Android and iOS platforms, this project focuses on Android as the target deployment platform owing to the limitation in the current development environment wherein the build process on iOS requires the use of macOS and Xcode, both of which are not available in the current Windows-based development environment.

2.4.2 Laravel Framework

Laravel is a web application development framework that is written in PHP language and implements the Model View Controller architectural paradigm. It consists of multiple built-in functionalities such as routing, authentication middleware, database support using Eloquent ORM, job scheduling, and broadcasting which enables real-time event communication. All of these built-in functionalities contribute to making development of applications easier and more efficient. In this project, Laravel is the main backend API server where the REST APIs that will be called by both Flutter app and web admin panel are hosted. Laravel framework is widely used for developing web-based information systems due to its structured development environment and various functionalities (Surahmat & Darmawan, 2024).

Recent literature has provided more proof on the appropriateness of Laravel as a framework for developing current-day web applications. According to Qabajeh et al. (2025), the architecture of Laravel follows the MVC pattern which means that the logic, interface, and data of the application are divided into separate entities, hence ensuring better organisation of the code and its scalability. Additionally, Qabajeh et al. noted that Laravel supports authentication, integration with databases, and RESTful APIs development and thus makes communication between front-end and back-end applications safe.

2.4.3 PostgreSQL Database

RDBMS is still an integral part of any modern information system because of the structured data storage, data integrity, and efficient query processing features. As it is possible to organize data in related tables in RDBMS, such a system provides the possibility to efficiently manage complex information and maintains data consistency. Haw et al. (2022) noted that relational databases are still able to ensure reliable performance for data-intensive applications. According to Akhtar (2023), PostgreSQL is consistently ranked among the most popular database management systems worldwide, reflecting its growing adoption among developers for building scalable and reliable applications.

Among the different RDBMS, PostgreSQL is one of the most widely used open-source databases because of its high reliability and good performance, as well as the ability to manage

large volumes of structured data. With its help, it becomes possible to store, organize, update, and retrieve data from it. Joiner et al. (2022) found out that PostgreSQL can be efficiently used in cloud-based solutions and perform well when it comes to transaction processing and data analysis. In this project, PostgreSQL is installed in the cloud on the platform called Neon. It means that PostgreSQL in this case works in the serverless environment with the dedicated cloud database having auto-scaling and auto-pause during the downtime.

2.4.4 Gemini API Integration

Gemini API is an artificial intelligence service provided by Google which is built upon large language models (LLMs) capable of creating text, offering context-aware writing suggestions, and helping users write structured content. According to Lang et al. (2024), large language models such as the Gemini family, including Gemini 1.5 Flash, are aimed at facilitating efficient language generation for various practical purposes. In this project, access to Gemini 1.5 Flash model is achieved with the help of Google AI Studio with the usage of a free-tier API key while all API requests are performed server-side in the Laravel backend controller to avoid exposing API key to client application. Integration of the system allows users to generate their logbook text with the help of AI, which means that the application will offer them writing suggestions, grammatical corrections, and structure improvement of their text.

2.4.5 Pusher Channels

Pusher Channels is a hosted real-time messaging service that facilitates two-way communication between the server and client applications via WebSocket connections. In the current project, Pusher Channels are used to facilitate real-time two-way communication between the students, lecturers, and company supervisors in the Laravel backend. Messages can be sent and displayed instantly in the application without the need for page refresh or reloading. Real-time communication systems based on WebSocket technology enhance responsiveness and user experience by facilitating instant exchange of data between clients and the server. With 200 simultaneous connections and 200,000 messages a day, the free plan offers more than enough for the needs of this academic project.

2.4.6 Firebase Cloud Messaging

Firebase Cloud Messaging (FCM) is a cross-platform push notification system created by Google, which, as of the date of writing, provides a service of delivering notifications from server-side apps to mobile devices without any costs and with no published limitation on usage. Firebase Cloud Messaging is a messaging service that provides a persistent connection between Google's servers and registered devices, thereby allowing receiving of notifications even when the targeted application is not working in foreground mode. In the current project, FCM is employed for sending of push notifications to students and supervisors for certain system events such as chat messages, logbook submissions, and feedback notifications. The payload of push notifications is being sent via the HTTP v1 API of FCM, while mobile application receives notifications through Firebase messaging services. Firebase Cloud Messaging has become very popular thanks to high scalability and reliability of message delivery for Android and iOS platforms without any requirements for infrastructure maintenance. Currently, in the free plan of FCM there is no any published limitation on the number of notifications and devices used; therefore, FCM can be regarded as an effective solution for push notifications for mobile apps.

2.4.7 Cloudinary

Cloudinary is a cloud-based solution that allows uploading, storing, delivering (via content delivery network (CDN)), and processing media files including images. Cloudinary is a media-management platform that can make the process of managing the media on web and mobile apps easier. In this project, Cloudinary is employed for the purpose of storing images that were uploaded by students using the mobile application. Laravel is used in order to upload files through the cloudinary-laravel SDK that can integrate into the Laravel filesystem abstraction and return CDN-backed URL in case the upload has been successfully completed. This URL is used by the mobile application in order to show the images more efficiently without saving the binary content in the main database. At the moment of writing, free tier offered by Cloudinary allows one to have up to 25 GB monthly of storage, bandwidth and processing, plus some URL-based transforms including resizing and compression, which can save the data volume consumed by mobile users. Please note that the numbers of the resources offered within the free tier may vary according to the provider's decision.

2.4.8 Laravel Sanctum and Role-Based Access Control

Sanctum is an out-of-the-box Laravel package that offers token-based authentication to mobile applications. The Sanctum system allows for each user to create their own API tokens to gain access to restricted routes. Following the success of the login attempt, the personal access token is generated by the Laravel backend and stored on the client's mobile device to be included as a Bearer token in all further API requests through the Authorization header of the HTTP request (Pari-Ayala & Ramos, 2025).

Role-based access control is implemented via the Spatie Laravel Permission package and enables the establishment of different user roles such as Student, Lecturer, Company Supervisor, and Administrator at the middleware, route, and controller levels. Using this approach allows for creating a robust structure for defining structural access boundaries through specific authorization and isolation workflow (Purba, 2026). This approach guarantees that only certain roles have access to certain system capabilities and data.

The Laravel Sanctum was chosen instead of third-party authentication providers since the package does not introduce any external dependencies, does not involve any additional costs, all credentials are saved in the same PostgreSQL database as the application itself, and is native to Laravel ecosystem.

2.5 Related Works and Similar Applications

2.5.1 A Digital Solution for Student Internship Tracking and Enhancing Student's Experience

Ahmed et al. (2026) introduced a model for the digital internships' tracking to facilitate internship management and communication between students and their supervisors. This research was based on certain problems in internship management, including poor communication, difficulties in monitoring the development of students, and absence of any internships' documentation. To solve these problems, Ahmed et al. created the digital platform allowing students to document their internships, submit reports, and communicate with their supervisors through the centralized system.

According to the research findings, the use of the digital system allowed improving internship monitoring, communication, and engagement during internship activities. Thanks to the digitization of internships' documentation, the process of documenting became much easier. Still, the system was created to monitor activities and manage internships, but there were no any intelligent tools such as AI writing, reminder notifications, and other features based on the use of images.

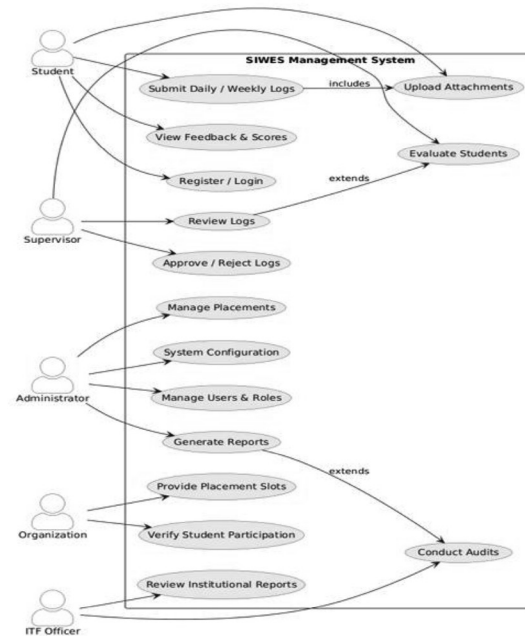


Figure 2.1 Proposed Student Internship Tracking Management System
(Source: Adapted from Ahmed et al. (2026))

2.5.2 The Development of an Integrated Cloud-Based System to Enhance Internship Management

E-Kehadiran is an integrated cloud-based internship management system that was developed by Wan Abdul Rahman et al. (2024). It was created with the goal of improving the process of attendance recording and logbook submission when running internship programs. The research problem for which E-Kehadiran was developed consisted of difficulties in managing attendance, recordkeeping, and communication between universities, internship supervisors, and interns. E-Kehadiran is a web-based system that incorporates such functions as user registration, attendance management, logbook submission, and performance monitoring.

As for implementation results, they were positive and showed that the system managed to increase the efficiency of administrative processes, recordkeeping, and internship monitoring. The user acceptance testing results were also positive and showed that users were satisfied with system integration, usability, and its ability to manage internship processes. However, this system is only aimed at helping with attendance management and logbooks and does not offer any intelligent features such as automated writing with the help of AI or automated feedback. Further development may include using GPS in attendance management and notifications about attendance recording.



Figure 2.2. Use case diagram of the E-Kehadiran system
(Source: Wan Abdul Rahman et al. (2024))

2.5.3 Mobile-Based Student Internship Monitoring System Using Progress Tracking Algorithm

The researcher Castro (2024) created a Mobile-based Student Internship Monitoring System that incorporates a Progress Tracking Algorithm to enhance the process of managing

and monitoring student internships. The research sought to solve problems related to manual internship monitoring, for instance, inaccurate attendance tracking, difficulty in locating student positions, and inefficient record management. The creation of the system involved creating an Android-based system that digitizes the student internship handbook and allows different users, including students, teachers, parents, partners, and administrators, to access and manage information related to internships. To facilitate accurate attendance tracking, the system is equipped with a GPS locator that confirms the positions of the students while clocking-in and out. The process of developing the system involved the use of Throw-Away Prototyping methodology, and UML diagrams were used in designing the system.

Evaluation of the system was carried out using the ISO 25010 software quality model with 30 participants comprising of student interns, teachers, parents, agencies partnering with the school, and IT professionals. Evaluation of the system yielded excellent results for functional suitability, reliability, performance efficiency, usability, security, compatibility, maintainability, and portability, meaning that the system improved internship monitoring and administrative process efficiency significantly. Besides, the system was capable of producing important reports like Daily Time Record (DTR) and task descriptions on a daily basis. Nevertheless, the system did not have any other additional features such as AI logbook writing and documentation. This means that future internship management systems should incorporate artificial intelligence to improve documentation and performance reporting processes.

2.5.4 Web-Based Student Internship Monitoring Information System

A Web-Based Student Internship Monitoring Information System was designed by Dewi and Hanifah (2024) for SMK Ma'arif NU 2 Boyolali, where the problems faced in the manual monitoring process of internships were solved. The previous system utilized journals, attendance lists, daily activity records, submission of final reports, and WhatsApp as the mode of communication. Such inefficiency in managing data and supervision led to the design of the new system, which was built using the SDLC approach and included three user types: administrator, students, and supervisors. In this system, students can take attendance, write their daily logbook, and upload the final report, whereas supervisors can check the same.

The developed system was tested using Black Box Testing and user acceptance tests conducted by the administrator, supervisor, and students. As a result, it turned out that all the system functionalities were performed perfectly, and the overall user satisfaction percentage was 87.53%, which means that the system helped to improve the internship process. Still, the system has not been equipped with any mobile features, real time tracking functionality, cloud integration and intelligent functionalities, such as AI-assisted documentation and analysis. Hence, some of the future improvements of the internship management system can include the integration of the mobile application and artificial intelligence.



Figure 2.3. Student Logbook Menu Page

(Source: Dewi and Hanifah (2024))

2.5.5 Comparison of Existing Systems

Several internship management systems discussed in Sections 2.5.1 to 2.5.4 demonstrate different approaches to improving internship monitoring, documentation, and communication. While each system offers valuable features, they differ in terms of functionality, accessibility, and level of technological advancement.

Table 2.1 Comparison between System

Features	Ahmed et al. (2026)	Wan Abdul Rahman et al. (2024)	Castro (2024)	Dewi & Hanifah (2024)	Proposed System
Digital Logbook	✓	✓	✓	✓	✓
Mobile Application	✓	✗	✓	✗	✓
Web-Based System	✓	✓	✗	✓	✓
Activity Recording	✓	✓	✓	✓	✓
Real-Time Monitoring	✓	✓	✓	Limited	✓
Attendance Tracking	Limited	✓	✓ (GPS-based)	✓	✓
Communication Features	✓	✓	Limited	Limited	✓
Cloud-Based Storage	✓	✓	✓	Limited	✓
AI-Assisted Writing	✗	✗	✗	✗	✓
Reminder Notifications	✗	✗	✗	✗	✓
Image Documentation	✗	✗	✗	✗	✓
Intelligent Feedback	✗	✗	✗	✗	✓

All four existing systems provide the core functions of digital logbooks, activity recording, and monitoring, but each has a different focus: Ahmed et al. (2026) and Wan Abdul Rahman et al. (2024) prioritize centralized platforms and communication, Castro (2024) emphasizes mobile and GPS-based tracking, while Dewi and Hanifah (2024) focus on web-based, structured supervision.

None of these systems, however, offer intelligent features such as AI-assisted writing, automated reminders, image documentation, or intelligent feedback. The proposed E-Logbook Mobile Application addresses this gap by combining these intelligent capabilities with the traditional monitoring functions, offering a more comprehensive solution for internship management.

Table 2.2 Summary Table for reviewed system

Title	Author, Year	Issues	Methods	Result	Limitation	Future Work
A Digital Solution for Student Internship Tracking and Enhancing Student's Experience	Ahmed et al., 2026	Traditional internship management lacks efficient communication, centralized monitoring, and intelligent assistance between students and supervisors.	Developed a digital internship tracking platform with activity recording, report submission, and centralized communication features.	Improved internship monitoring, communication efficiency, and student engagement through a centralized system.	Focused mainly on tracking and management without AI-assisted writing, automation, or advanced intelligent features.	Future research may include AI-assisted writing, automated reminders, and intelligent documentation support.
The Development of an Integrated Cloud-Based System to Enhance Internship Management	Wan Abdul Rahman et al., 2024	Difficulty in tracking attendance, maintaining accurate records, and ensuring effective communication among stakeholders.	Developed a cloud-based system (E-Kehadiran) with attendance tracking, logbook submission, and centralized monitoring.	Improved administrative efficiency, data accessibility, and communication	Lacked intelligent features such as AI-assisted writing, automated feedback, and smart notification systems.	Future improvements may include GPS-based attendance, notification reminders, and intelligent feedback features.

Title	Author, Year	Issues	Methods	Result	Limitation	Future Work
Mobile-Based Student Internship Monitoring System Using Progress Tracking Algorithm	Castro, 2024	Existing systems lack accurate attendance tracking, real-time monitoring, and efficient record management.	Developed a mobile-based internship system integrated with GPS tracking and progress tracking algorithms.	Improved attendance accuracy, real-time monitoring, and internship activity tracking efficiency.	Focused mainly on monitoring and tracking without AI-based documentation, analytics dashboards, or writing assistance.	Future work could integrate AI-assisted documentation, analytics dashboards, and enhanced reporting tools.
Web-Based Student Internship Monitoring Information System	Dewi & Hanifah, 2024	Manual systems using paper-based records and messaging apps cause inefficient monitoring and data management.	Developed a web-based system using SDLC with features such as attendance, logbook submission, and supervisor verification.	Improved internship monitoring, documentation management, and achieved high user satisfaction.	Limited to web platform without mobile access, real-time tracking, cloud integration, or intelligent automation.	Future enhancements may include mobile applications, cloud integration, real-time tracking, and AI-based features.